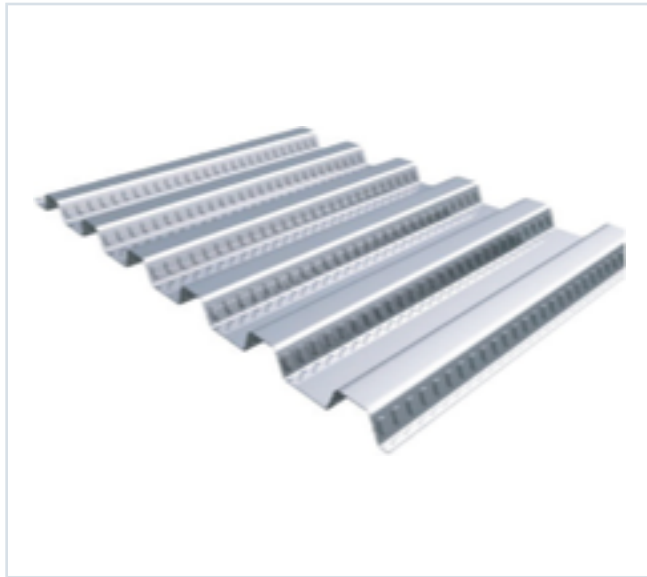




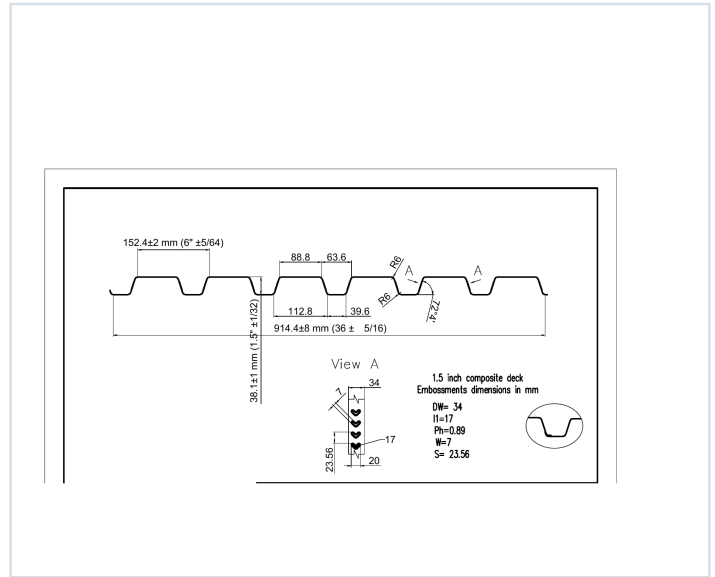
DECKING PRODUCT SUBMITTAL

PRODUCT	GRADE	GAUGE	COATING
1.5" Composite Deck	Grade 50	18 ga	G90

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in³/ft)	S- (in³/ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in■/ft)	I- (in■/ft)
0.0474	2.6	50	0.294	0.310	8812	9291	0.257	0.290

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 45, 46, 47, 48
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE

15CD-FY50-18-G90 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 45

GRADE 50 STEEL

View A

Section Properties

Gage	Design Thickness (inches)	Weight (lb/ft)	F _y (ksi)	F _u (ksi)	S _x (in ³)	S _y (in ³)	ASD (3 + 147) M _u /Q (in ³)	ASD (3 + 147) M _u /Q (in ³)	I _x (in ⁴)	I _y (in ⁴)
22	0.0295	16	50	0.770	0.179	0.021	558	558	0.144	0.107
20	0.0368	20	50	0.246	0.222	6407	6461	0.182	0.290	
18	0.0474	26	50	0.294	0.330	880	908	0.257	0.290	
16	0.0598	33	50	0.378	0.390	1327	1467	0.341	0.363	

Note: All section properties and ASD/beam strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10 and AISC 360-10.

Shear and Web Crippling

Gage	V _u /Q (lb/ft)	Web Crippling (R _u /Q) (lb/ft) One Flange Loading End Bearing	Web Crippling (R _u /Q) (lb/ft) Two Flange Loading Interior Bearing	
22	2424	403	1015	1096
20	3803	1643	1535	1605
18	5052	194	2097	2773
16	6279	205	3183	3606

Note: All section properties and ASD/beam strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10 and AISC 360-10.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
Single	22	158	12	94	80	69	60	53	47	42	38	34
	20	272	142	100	102	88	77	67	60	53	48	43
	18	235	194	163	139	120	104	92	81	73	65	59
	16	202	250	207	179	154	134	119	106	95	86	78
Double	22	143	118	89	85	75	64	56	49	44	40	36
	20	198	147	125	105	91	79	69	61	55	49	44
	18	246	205	172	147	126	110	97	86	78	69	62
	16	311	257	216	184	159	138	122	106	96	86	78
Triple	22	179	148	124	106	91	79	70	62	55	49	45
	20	262	183	154	131	113	99	87	77	69	62	56
	18	310	256	215	183	158	138	121	107	96	86	77
	16	389	321	270	230	198	173	152	135	120	108	97

STEEL CATALOG PAGE 46

Span

Gage

5'-0"

5'-6"

6'-0"

6'-6"

7'-0"

7'-6"

8'-0"

8'-6"

9'-0"

9'-6"

10'-0"

Single

22

16

20

18

16

Double

22

20

18

16

Triple

22

20

18

16

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.607.

Construction Span Table - 20 psf Construction Load

Total Slab Depth	Deck Type	Normal Weight Concrete (145 pcf)			Lightweight Concrete (115 pcf)			
		Maximum Unstressed Clear Span	1 span	2 span	3 span	Maximum Unstressed Clear Span	1 span	2 span
3'-0" (1'-0") 33 PSF	15x16x22 ga	6'-0"	7'-0"	7'-9"	6'-0"	7'-0"	7'-9"	8'-0"
	15x16x18 ga	7'-0"	8'-0"	8'-9"	7'-0"	8'-0"	8'-9"	9'-0"
	15x16x16 ga	8'-0"	9'-0"	9'-9"	8'-0"	9'-0"	9'-9"	10'-0"
	15x16x14 ga	9'-0"	10'-0"	10'-9"	9'-0"	10'-0"	10'-9"	11'-0"
4'-0" (1'-250) 37 PSF	15x16x22 ga	6'-0"	7'-0"	7'-9"	6'-0"	7'-0"	7'-9"	8'-0"
	15x16x18 ga	7'-0"	8'-0"	8'-9"	7'-0"	8'-0"	8'-9"	9'-0"
	15x16x16 ga	8'-0"	9'-0"	9'-9"	8'-0"	9'-0"	9'-9"	10'-0"
	15x16x14 ga	9'-0"	10'-0"	10'-9"	9'-0"	10'-0"	10'-9"	11'-0"
4'-0" (1'-400) 43 PSF	15x16x22 ga	5'-0"	6'-0"	7'-0"	5'-0"	6'-0"	7'-0"	8'-0"
	15x16x18 ga	6'-0"	7'-0"	8'-0"	6'-0"	7'-0"	8'-0"	9'-0"
	15x16x16 ga	7'-0"	8'-0"	9'-0"	7'-0"	8'-0"	9'-0"	10'-0"
	15x16x14 ga	8'-0"	9'-0"	10'-0"	8'-0"	9'-0"	10'-0"	11'-0"
5'-0" (1'-550) 49 PSF	15x16x22 ga	4'-0"	5'-0"	6'-0"	4'-0"	5'-0"	6'-0"	7'-0"
	15x16x18 ga	5'-0"	6'-0"	7'-0"	5'-0"	6'-0"	7'-0"	8'-0"
	15x16x16 ga	6'-0"	7'-0"	8'-0"	6'-0"	7'-0"	8'-0"	9'-0"
	15x16x14 ga	7'-0"	8'-0"	9'-0"	7'-0"	8'-0"	9'-0"	10'-0"
5'-0" (1'-400) 43 PSF	15x16x22 ga	4'-0"	5'-0"	6'-0"	4'-0"	5'-0"	6'-0"	7'-0"
	15x16x18 ga	5'-0"	6'-0"	7'-0"	5'-0"	6'-0"	7'-0"	8'-0"
	15x16x16 ga	6'-0"	7'-0"	8'-0"	6'-0"	7'-0"	8'-0"	9'-0"
	15x16x14 ga	7'-0"	8'-0"	9'-0"	7'-0"	8'-0"	9'-0"	10'-0"
6'-0" (1'-400) 43 PSF	15x16x22 ga	3'-0"	4'-0"	5'-0"	3'-0"	4'-0"	5'-0"	6'-0"
	15x16x18 ga	4'-0"	5'-0"	6'-0"	4'-0"	5'-0"	6'-0"	7'-0"
	15x16x16 ga	5'-0"	6'-0"	7'-0"	5'-0"	6'-0"	7'-0"	8'-0"
	15x16x14 ga	6'-0"	7'-0"	8'-0"	6'-0"	7'-0"	8'-0"	9'-0"

Note: Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

15CD-FY50-18-G90 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 47

LSC Deck - Wide Flange Up				ISNV Deck - Wide Flange Down			
50 ksi				50 ksi			
Live Load	20	psf	with concrete	Live Load	20	psf	with concrete
DL	160	psf		DL	160	psf	
Deck Weight	2	psf		Deck Weight	2	psf	
Live Load	50	psf	without concrete	Live Load	50	psf	without concrete
Total Slab Thickness	Deck Gauge	Maximum Cantilever (inches) NWC	Maximum Cantilever (inches) LWC	Total Slab Thickness	Deck Gauge	Maximum Cantilever (inches) NWC	Maximum Cantilever (inches) LWC
3.5	22	36	36	3.5	22	25	25
3.5	20	31	31	3.5	20	30	30
3.5	18	40	40	3.5	18	38	38
3.5	16	47	47	3.5	16	46	46
4	22	36	36	4	22	25	25
4	20	31	31	4	20	30	30
4	18	40	40	4	18	38	38
4	16	47	47	4	16	46	46
4.5	22	36	36	4.5	22	25	25
4.5	20	31	31	4.5	20	30	30
4.5	18	40	40	4.5	18	38	38
4.5	16	47	47	4.5	16	46	46
5	22	36	36	5	22	25	25
5	20	31	31	5	20	30	30
5	18	40	40	5	18	38	38
5	16	47	47	5	16	46	46
5.5	22	36	36	5.5	22	25	25
5.5	20	31	31	5.5	20	30	30
5.5	18	40	40	5.5	18	38	38
5.5	16	47	47	5.5	16	46	46
6	22	36	36	6	22	25	25
6	20	31	31	6	20	30	30
6	18	40	40	6	18	38	38
6	16	47	47	6	16	46	46

— Assumes that backspan is greater than the cantilever spans.
— Applicable to single, double, triple spans or greater.
— Controlled by concentrated load = 30 psf.
— Controlled by concrete weight = 10 psf.
— Controlled by Deflection.

STEEL CATALOG PAGE 48

22 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	31	400	400	372	334	289	252	210	170
4	37	400	400	400	397	340	298	258	218
4.5	43	400	400	400	400	400	397	331	281
5	49	400	400	400	400	400	400	369	319
5.5	55	400	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	277	156	135	121	110	96	85	80
4	224	197	175	156	139	125	113	102
4.5	273	241	213	190	170	153	138	126
5	323	286	253	226	200	182	164	149
5.5	375	331	294	262	235	211	191	173
6	400	378	336	300	269	242	218	197

20/18/16 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	31	400	400	400	400	400	400	400	400
4	37	400	400	400	400	400	400	400	400
4.5	43	400	400	400	400	400	400	400	400
5	49	400	400	400	400	400	400	400	400
5.5	55	400	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	236	181	170	151	136	122	110	100
4	274	242	215	190	172	155	140	127
4.5	335	296	263	235	215	195	172	156
5	398	362	315	280	251	226	203	186
5.5	400	400	364	325	292	263	238	216
6	400	400	400	372	334	301	272	247

Note
Due to the profile of the embossments, there is no gain in strength for the composite deck-slab when the deck gets thicker than 20 gage. However, the construction square do get longer for 18 and 16 gage deck.